

AD EXP-9

Name: Abhinit Khambekar

Roll No.: 42434

Batch: S-5

Code:

JAVA CODE:

MainActivity.java:

```
package com.example.datastorageapp;

import android.content.Context;
import android.os.Bundle;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;
import java.io.*;

public class MainActivity extends AppCompatActivity {

    EditText etFileName, etData;
    RadioButton rbInternal, rbExternal;
    Button btnSave, btnLoad;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        etFileName = findViewById(R.id.etFileName);
        etData = findViewById(R.id.etData);
        rbInternal = findViewById(R.id.rbInternal);
        rbExternal = findViewById(R.id.rbExternal);
        btnSave = findViewById(R.id.btnSave);
        btnLoad = findViewById(R.id.btnLoad);

        btnSave.setOnClickListener(v -> saveData());
        btnLoad.setOnClickListener(v -> loadData());
    }

    private void saveData() {
        String fileName = etFileName.getText().toString().trim();
        String data = etData.getText().toString().trim();

        if(fileName.isEmpty() || data.isEmpty()) {
            Toast.makeText(this, "Enter filename and data", Toast.LENGTH_SHORT).show();
            return;
        }
    }
}
```

```

try {
    if(rbInternal.isChecked()) {
        FileOutputStream fos = openFileOutput(fileName, Context.MODE_PRIVATE);
        fos.write(data.getBytes());
        fos.close();
        Toast.makeText(this, "Saved to Internal Storage",
Toast.LENGTH_SHORT).show();
    }
    else if(rbExternal.isChecked()) {
        File file = new File(getExternalFilesDir(null), fileName);
        FileOutputStream fos = new FileOutputStream(file);
        fos.write(data.getBytes());
        fos.close();
        Toast.makeText(this, "Saved to External Storage",
Toast.LENGTH_SHORT).show();
    }
    else {
        Toast.makeText(this, "Select Storage Type", Toast.LENGTH_SHORT).show();
    }

} catch (Exception e) {
    Toast.makeText(this, "Error: " + e.getMessage(), Toast.LENGTH_SHORT).show();
}
}

```

```

private void loadData() {
    String fileName = etFileName.getText().toString().trim();

    if(fileName.isEmpty()) {
        Toast.makeText(this, "Enter filename", Toast.LENGTH_SHORT).show();
        return;
    }

    try {
        BufferedReader reader;
        if(rbInternal.isChecked()) {
            FileInputStream fis = openFileInput(fileName);
            reader = new BufferedReader(new InputStreamReader(fis));
            Toast.makeText(this, "Loaded from Internal Storage",
Toast.LENGTH_SHORT).show();
        }
        else if(rbExternal.isChecked()) {
            File file = new File(getExternalFilesDir(null), fileName);
            FileInputStream fis = new FileInputStream(file);
            reader = new BufferedReader(new InputStreamReader(fis));
            Toast.makeText(this, "Loaded from External Storage",
Toast.LENGTH_SHORT).show();
        }
        else {

```

```

        Toast.makeText(this, "Select Storage Type", Toast.LENGTH_SHORT).show();
        return;
    }

    StringBuilder sb = new StringBuilder();
    String line;
    while((line = reader.readLine()) != null){
        sb.append(line).append("\n");
    }
    reader.close();
    etData.setText(sb.toString().trim());

    } catch (Exception e) {
        Toast.makeText(this, "Error: " + e.getMessage(), Toast.LENGTH_SHORT).show();
    }
}
}

```

XML CODE:

activity_main.xml:

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:padding="20dp"
    android:background="#F0F4F8"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <TextView
        android:text="Data Storage App"
        android:textSize="22sp"
        android:textStyle="bold"
        android:gravity="center"
        android:layout_marginBottom="20dp"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <EditText
        android:id="@+id/etFileName"
        android:hint="Enter Filename (e.g. aman.txt)"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <EditText
        android:id="@+id/etData"
        android:hint="Enter Data"
        android:gravity="top"

```

```
android:minHeight="150dp"
android:layout_marginTop="10dp"
android:layout_width="match_parent"
android:layout_height="wrap_content"/>
```

```
<RadioGroup
    android:id="@+id/storageGroup"
    android:orientation="horizontal"
    android:layout_marginTop="10dp"
    android:layout_width="match_parent"
    android:layout_height="wrap_content">
```

```
<RadioButton
    android:id="@+id/rbInternal"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Internal Storage"/>
```

```
<RadioButton
    android:id="@+id/rbExternal"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="External Storage"/>
```

```
</RadioGroup>
```

```
<LinearLayout
    android:layout_marginTop="15dp"
    android:gravity="center"
    android:layout_width="match_parent"
    android:layout_height="wrap_content">
```

```
<Button
    android:id="@+id/btnSave"
    android:text="SAVE"
    android:layout_marginEnd="10dp"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"/>
```

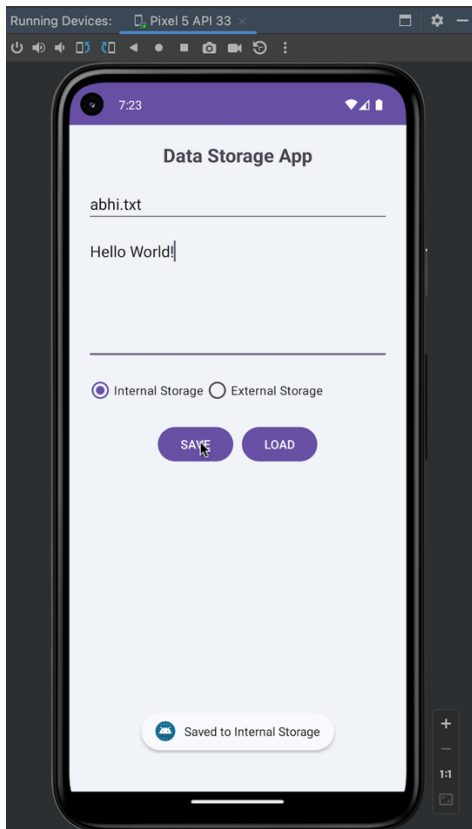
```
<Button
    android:id="@+id/btnLoad"
    android:text="LOAD"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"/>
```

```
</LinearLayout>
```

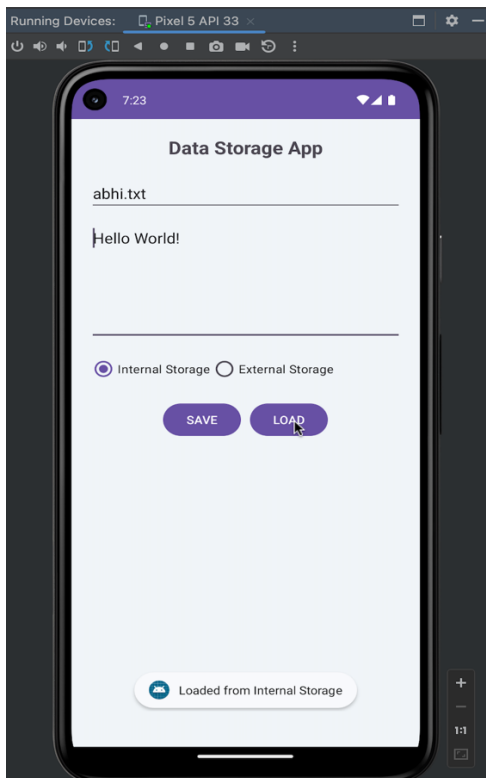
```
</LinearLayout>
```

O/P:

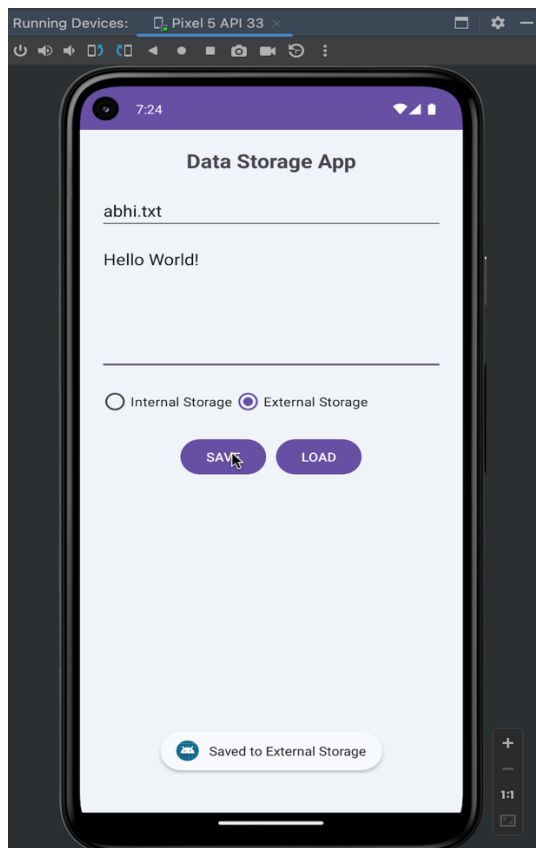
1.) Saving to Internal Storage



2.) Loading from Internal Storage



3.) Saving to External Storage



4.) Loading from External Storage

